

# Tridev Patel

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## PROFESSIONAL SUMMARY

Electronics and Communication Engineering undergraduate with hands-on experience in embedded systems development at DRDO. Proficient in Embedded C, SPI, I2C, and UART communication protocols with expertise in register-level programming, firmware development, and microcontroller interfacing. Actively upskilling in Artificial Intelligence, Machine Learning, and Generative AI. IEEE member with a strong foundation in both hardware and software engineering.

## EDUCATION

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|---------------------------------------------------------------|----------------------|
| <b>Guru Tegh Bahadur 4th Centenary Engg. College (GGSIPU)</b> | Delhi, India         |
| <b>B.Tech in Electronics and Communication Engineering</b>    | 2023 – 2027          |
| <b>UP Board of Secondary Education</b>                        | Uttar Pradesh, India |
| <b>12th Class (Science) — 73%</b>                             | 2022                 |
| <b>UP Board of Secondary Education</b>                        | Uttar Pradesh, India |
| <b>10th Class — 79%</b>                                       | 2020                 |

## EXPERIENCE

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| <b>Embedded Systems Intern</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                      | 2025 |
| <i>Defence Research and Development Organisation (DRDO) – SAG, New Delhi, India</i>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                 |      |
| <ul style="list-style-type: none"><li>Developed low-level Embedded C drivers for SPI, I2C, and UART communication protocols on ARM-based microcontrollers.</li><li>Performed register-level programming to directly configure peripheral control registers, eliminating HAL abstraction overhead.</li><li>Configured baud rate, clock polarity/phase (CPOL/CPHA), data frame format, and chip-select logic for multi-slave SPI buses.</li><li>Implemented both polling and interrupt-driven TX/RX mechanisms, ensuring deterministic real-time data transmission.</li><li>Debugged serial communication errors using logic analysers and UART serial monitors; resolved timing and synchronisation faults.</li><li>Validated real-time data integrity between sensors, EEPROM, and display peripherals in an embedded system prototype.</li><li>Authored driver API documentation and firmware test reports for the SAG engineering team.</li></ul> |      |

## PROJECTS

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| <b>Embedded Communication Driver Suite</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                  | 2025 – Present |
| <i>Embedded C · SPI · I2C · UART · ARM Cortex-M · Git</i>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                   |                |
| <ul style="list-style-type: none"><li>Designed a modular firmware library for SPI, I2C, and UART protocols on ARM Cortex-M MCUs with full register-level control.</li><li>Implemented interrupt-driven circular FIFO buffers for non-blocking RX/TX, achieving reliable throughput up to 1 Mbps.</li><li>Validated multi-device communication (sensor arrays, EEPROM, OLED displays); resolved timing violations via logic-analyser captures.</li><li>Maintained clean, modular codebase with Git version control and structured hardware bring-up documentation.</li></ul> |                |
| <b>Responsive Web Portfolio</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                             | 2025           |
| <i>HTML5 · CSS3 · JavaScript (ES6+) · Git · GitHub Pages</i>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                |                |
| <ul style="list-style-type: none"><li>Built a mobile-first responsive portfolio using semantic HTML5, CSS Flexbox/Grid, and vanilla JavaScript ES6+.</li><li>Ensured cross-browser compatibility and WCAG-compliant colour contrast; deployed via GitHub Pages with Git-based CI workflow.</li></ul>                                                                                                                                                                                                                                                                        |                |

## CERTIFICATIONS & TRAINING

- Introduction to Generative AI** — Google | Generative AI fundamentals, diffusion models, applications in automation and content generation.
- Introduction to Large Language Models** — Google | Transformer architecture, self-attention, fine-tuning, and prompt engineering.
- Planning a Machine Learning Project** — Google / Coursera | ML project lifecycle, data preprocessing, model selection, and evaluation metrics.

- **C Programming** — NPTEL / Online | Memory management, pointers, structures, and algorithm design in ANSI C.
- **Web Development Fundamentals** — Online | HTML5, CSS3, JavaScript ES6+, responsive design, and Flexbox/Grid layout.

**TECHNICAL SKILLS**

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|--------------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>Programming Languages</b>   | Embedded C · C · Python · JavaScript (ES6+) · HTML5 · CSS3                                                                                                    |
| <b>Communication Protocols</b> | SPI · I2C · UART · Serial Communication · CAN (Exposure) · Real-Time Data Transmission                                                                        |
| <b>Embedded &amp; Hardware</b> | ARM Cortex-M · Microcontroller Interfacing · Register-Level Programming · Interrupt Handling · Polling · Firmware Development · Logic Analyser · Oscilloscope |
| <b>Developer Tools</b>         | Git · GitHub · VS Code · Embedded Debuggers · GitHub Pages                                                                                                    |
| <b>AI / ML Technologies</b>    | Generative AI · Large Language Models (LLM) · Prompt Engineering · Machine Learning · Neural Networks · Data Preprocessing                                    |
| <b>Soft Skills</b>             | Technical Documentation · Problem Solving · Team Collaboration · Research & Analysis · IEEE Community                                                         |

**ACHIEVEMENTS & ACTIVITIES**

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- Completed internship at DRDO – SAG (Scientific Analysis Group), one of India's premier defence R&D organisations.
- Successfully implemented embedded communication drivers deployed in a real DRDO prototype system.
- Active IEEE member — engaged in emerging technology discussions, workshops, and networking events.
- Completed 5+ certifications in AI, ML, C Programming, and Web Development from Google and leading platforms.
- Continuously self-upskilling in Embedded Systems, AI/ML, and Full-Stack Web Development.